Higher National Diploma in Civil Engineering

ThirdYear, Second Semester Examination – 2022

CE 3118: Management of Civil Engineering Construction

Instructions for candidates:	No. of questions	: 06
Answer five questions only.	No. of pages	: 02
All questions carry equal marks.	Time	: 03 Hours

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01. i. Define the term "Project Management". (2 Marks)
    ii. List out any fourqualities of a Project Manager. (4 Marks)
    iii. Briefly describe "Triple Constraints" of project management (6 Marks)
    iv. Write down and explainat least four types of types of building construction systems commonly found in the world today. (8 Marks)

(Total 20 Marks)

02. i. What do you mean by the term "Authority and Responsibility" (2 Marks)
    ii. State four importance of Human Resource Management. (4 Marks)
    iii. Differentiate the terms "Efficiency and Effectiveness". (6 Marks)
    iv. Describe the levels of management with relevant example. (8 Marks)

(Total 20 Marks)

03. i. What did you understand about "Law of Supply" (2 Marks)
    ii. Mention four users of "Accounting information". (4 Marks)
    iii. The material X is used uniformly throughout the year. The data about annual requirement, ordering cost and holding cost of this material is given below:
    Annual requirement: 12250 units
    Ordering cost: Rs10 per order
    Holding cost: Rs 0.50 per unit
    You are required to calculate the economic order quantity (EOQ) of material X using above data.
    (6 Marks)
    iv. In a hypothetical market, the demand is given by QD = 200 –P and supply is given by QS = 100+4P. Calculate the Market equilibrium price and quantity.
    (8 Marks)

(Total 20 Marks)

04. i. What do you mean by "Law of Contract " in Sri Lanka (2 Marks)

ii. State major components to be accomplish to form a contract. (4 Marks)

iii. Briefly explain "key performance Indicators" with example (6 Marks)

iv. "Cash flow forecasting helps to manage the cash flows on construction projects" Do you agree with this statement? Explain. (8 Marks)

(Total 20 Marks)

05. i. What is the significance of "Network Analysis" technique? (2 Marks)

ii. Differentiate the terms "Float and Total Float" in the network Analysis.(4 Marks)

iii. The following set of activities and precedence relationships have been developed with respect to a project.

Activity	Immediate Predecessors	Duration (Days)
A	-	7
B	-	3
C	A	3
D	A	8
E	A	3
F	C	3
G	D	4
H	B, E	6
I	H	2
J	F, G, I	2

a) Draw the network diagram (4Marks)

b) Identify the Critical path and Project completion time (4 Marks)

c) Calculate Early Finished Time (EFT), Late Finished Time (LFT), and Total Float (TF) for each activity. (6 Marks)

(Total 20 Marks)

06. i. Define the term "Site Storage" (2 Marks)

ii. List out four responsibility of a store keeper. (4 Marks)

iii. "A hazard is anything that has the potential to harm the health and safety of a person"

d) Give six main hazards that are encountered on a typical construction site. (6 Marks)

e) Explain four ways that helpsto construction businesses to reduce the workplace accidents and promote construction site safety. (8 Marks)

(Total 20 Marks)